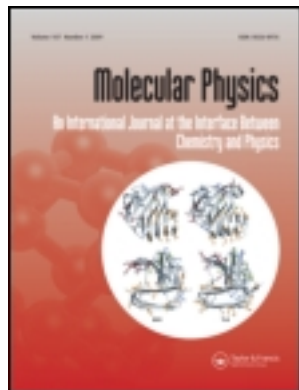


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A rescaled MSA structure factor for dilute charged colloidal dispersions

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The structure of a dispersion of charged colloidal particles interacting through a screened Coulomb potential is well described at moderate to high densities by calculating the correlations in the mean spherical approximation (MSA). In this paper we extend the validity of the MSA calculation to arbitrarily low densities, using a physical rescaling argument which preserves the analytic form of the MSA solution. Our results are in excellent agreement with other numerical calculations and with experimental low-density light scattering data. The method may be viewed as a generalization of Gillan's prescription for one component plasmas to systems interacting through a Yukawa potential. An advantage of the present prescription is that it allows a smooth transition from strong to weak coupling, and it implies no functional relation between the experimentally independent parameters.

Aqueous dispersions or solutions of charged micelles or colloidal particles exhibit a considerable amount of short range order, evident from light or small-angle neutron scattering measurements of their static structure factor [1, 2]. An analytical solution of the mean spherical approximation (MSA) for a simple model of charged particle dispersions has recently been proposed [3] and successfully applied to an analysis of the structure of concentrated micellar solutions [2]. In the present paper, this theoretical model is extended to the study of the static structure of dilute dispersions of highly charged colloids, using a rescaling argument based on the generalization of a procedure due to Gillan [4].

Since the diameter of colloidal particles is typically of the order of 10^2 – 10^3 Å, the solvent is adequately modelled by a continuum of dielectric constant ϵ , while the counter ions and any added electrolyte will be characterized by their Debye screening length λ_D . Neglecting the weak van der Waals attraction, we model the Coulomb repulsion between spherical particles by the standard potential [5] due to the mutual interaction of their electric double-layers:

$$v(r) = \pi\epsilon_0\epsilon\sigma^2\psi_0^2 \exp[-\kappa(r-\sigma)]/r, \quad r > \sigma, \quad (1)$$

where σ is the particle diameter, ψ_0 is the surface potential and $\kappa = 1/\lambda_D$. Choosing σ as the unit of length, the reduced potential reads

$$\beta v(x) = \gamma \exp(-kx)/x, \quad x > 1, \quad (2)$$

where $x = r/\sigma$, $k = \kappa\sigma$, $\beta = 1/k_B T$ and γ is the dimensionless coupling constant

$$\gamma = \beta\pi\epsilon_0\epsilon\sigma\psi_0^2 \exp(k) \quad (3)$$

If n is the number concentration of particles, the suspension is entirely characterized by γ , k and the volume fraction $\eta = \pi n \sigma^3 / 6$. An alternative choice for the length scale is the mean interparticle spacing $a = (3/4\pi n)^{1/3}$, and the corresponding Coulomb coupling constant is conveniently defined as

$$\Gamma_k = 2\beta v(r=2a). \quad (4)$$

Note that $a/\sigma = 1/(2\eta^{1/3})$; in the absence of screening ($k \rightarrow 0$), Γ_k coincides with the usual coupling constant of the one component plasma (OCP) [6].

The standard MSA closure reads [7]

$$\left. \begin{aligned} g(r) &= 0, & r < \sigma, \\ c(r) &= -\beta v(r), & r > \sigma, \end{aligned} \right\} \quad (5)$$

where $g(r)$ and $c(r)$ are the pair distribution and direct correlation functions. Together with the Ornstein-Zernike relation [7], equations (5) form a closed set. In the case of the Yukawa potential (2), analytic solutions have been obtained by Waisman [8], and in a somewhat different form by Pastore *et al.* [9] and by Hayter and Penfold [3]; we have verified that the solutions are entirely equivalent, but the Hayter-Penfold (HP) version is much more directly evaluated, and has been adopted here. In the limit of no screening ($k=0$), an analytic solution of the MSA had been previously obtained by Palmer and Weeks [10].

In colloidal systems the Coulomb coupling is often large, i.e. γ (or Γ_k) $\gg 1$. Under such conditions the MSA is known to yield reliable structure factors only for sufficiently high concentrations ($\eta \gtrsim 0.2$); this is precisely the case of the concentrated micellar solutions studied in [2]. In the case of charged colloidal particles such as polystyrene spheres [1], the Coulomb coupling is generally strong, but packing fractions are typically of the order of 10^{-3} or less. Under these conditions the MSA is of no direct use; in particular it yields unphysical negative contact values, $g(r=\sigma^+)$, of the pair distribution function.

We overcome this difficulty by noting that in systems where $\gamma \gg 1$, the value of the potential (1) at contact is much larger than $k_B T$, and contact configurations are extremely unlikely. This means that $g(r)$ is practically zero for $r \lesssim a$, due to Coulomb correlations alone [6], and the hard core plays no physical rôle. We thus adopt the spirit of the idea which Gillan [4] applied to describe the OCP (where $\sigma=0$, $k=0$), by increasing the particle diameter from its physical value σ to an effective hard core value σ' , while maintaining the Coulomb coupling Γ_k constant. Since fixing the density fixes a , it is clear from (2) and (4) that this amounts to calculating the structure for a rescaled potential with $\gamma' = \gamma s$ and $k' = k/s$, where $s = \sigma/\sigma'$. As long as the effective contact potential $v(r=\sigma')$ remains large relative to $k_B T$, the physics of the system is unaffected by this procedure, but since the effective packing fraction $\eta' = \pi n \sigma'^3 / 6$ increases rapidly with σ' , the MSA provides an increasingly more accurate description of the system. The structure factor $S(\eta, \gamma, k; K)$ of the physical system may thus be accurately calculated by evaluating $S(\eta', \gamma', k'; K')$ for the rescaled system, and then trivially rescaling the result by a contraction of the reduced wave-numbers $K' = Q\sigma'$ to the relevant scale $K = Q\sigma = K's$, where Q is the momentum transfer.

There remains the problem of how to determine σ' for the rescaled MSA (RMSA) calculation. To this end, we have generalized the Gillan criterion [4] for the determination of the effective diameter σ' as follows. We begin by calculating the MSA contact value $g_\eta(r=\sigma^+)$ at the physical packing fraction η . In the notation of HP this can be cast in the form

$$g_\eta^{\text{MSA}}(r=\sigma^+) = -(p_1 + p_2 C + p_3 F), \quad (6)$$

where all coefficients are defined in terms of η , γ and k in [3]. If this contact value turns out to be non-negative, we simply choose $\sigma' = \sigma$, and no rescaling of the MSA solution is necessary; this occurs whenever the packing fraction is large enough for the hard core to play a physically significant rôle. On the other hand, if (as is usually the case for colloids) the contact value (6) is negative, signalling the predominance of Coulomb correlations and the failure of the MSA, σ' is increased from σ until the condition

$$g_{\eta'}^{\text{RMSA}}(r=\sigma'^+) = 0 \quad (7)$$

is satisfied, where RMSA refers to the MSA calculated for the rescaled potential. Combining (6) and (7) with the equations of [3] for the coefficients appearing in (6), an algebraic equation is derived for η' , which is easily solved by a standard iterative Newton algorithm for given values of γ (or Γ_k) and k ; the potential must be rescaled at fixed coupling, as described above, at each step in the iteration, which is rapidly convergent.

We now have a clear prescription for calculating the structure factor of colloid suspensions of any concentration, on the basis of the potential model (1) and the MSA solution of reference [3]. In particular, this RMSA procedure overcomes the two limitations inherent in Gillan's original prescription for the OCP [4]; the present method allows a smooth transition from strong to weak (or zero) coupling, and it permits the independent experimental parameters η , γ and k to remain independent.

In the special case of the OCP ($\sigma=0$, $k=0$), the procedure has already been tested by Gillan [4] who found good agreement with 'exact' Monte Carlo data for the pair distribution function [11]. The MSA also does reasonably well for charged hard spheres ($\sigma \neq 0$, $k=0$) at high packing fractions [12] or very low coupling, where no rescaling is necessary. In order to provide a systematic check of our procedure under conditions typical of the colloidal suspensions ($\eta \lesssim 10^{-3}$, $k \gtrsim 0.1$, $\gamma \gg 1$) which are our present concern, we have obtained extensive numerical solutions of the hypernetted chain (HNC) equation [7], which is known to yield reasonably accurate structure factors in the case of long range coulombic interactions [6]. The agreement between the RMSA and the HNC results for $S(K)$ is outstanding, as illustrated in figure 1; the computing time for the two cases differs by more than two orders of magnitude in favour of the RMSA [13].

For very strong Coulomb couplings, HNC theory tends to become less accurate. To overcome its defects, Rosenfeld and Ashcroft [14] proposed a semi-empirical modification of the HNC equation (MHNC) which takes into account the contribution of the missing 'bridge' diagrams on the basis of an effective hard sphere bridge function. We have tested this scheme against available Monte Carlo data for charged hard spheres ($k=0$) [12], and then extended the calculations to the case of finite screening ($k>0$); we conclude

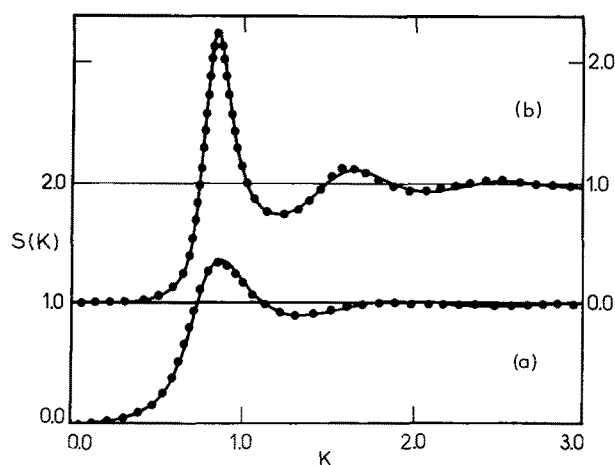


Figure 1. Comparison of $S(K)$ vs. $K=Q\sigma$ calculated by RMSA (—) and HNC (●) for $\eta=10^{-3}$, $k=\kappa\sigma=0.2$; (a) $\gamma=200$ ($\Gamma_k=5.4$), (b) $\gamma=1000$ ($\Gamma_k=27.1$).

that the RMSA and HNC results somewhat underestimate the structure at low density in the limit of very strong Coulomb coupling. In particular, although the main peak in $S(K)$ is predicted to be at the correct position, its amplitude may be underestimated by as much as 20 per cent in the situation corresponding to figure 1(b).

The aforementioned defects being significant only under rather extreme conditions, the RMSA scheme provides a convenient tool for analysing experimental data on colloidal dispersions over a wide range of concentrations and ionic strengths. In such experiments the colloid and electrolyte concentrations are fairly well controlled parameters, but the surface potential ψ_0 can be only rather poorly estimated. As an example, we consider the well known data of Brown

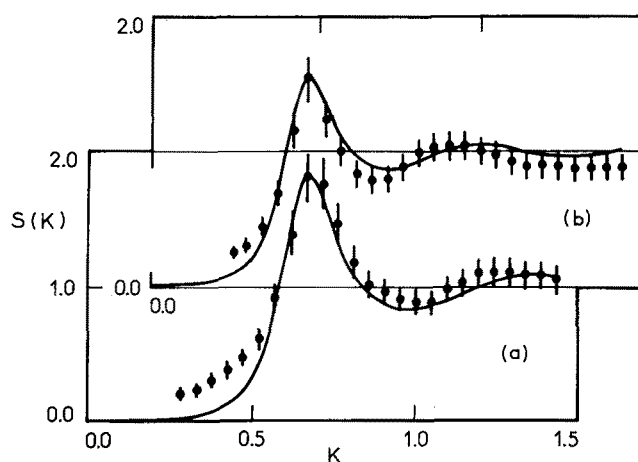


Figure 2. Comparison between RMSA (—) and data of Brown *et al.* [1] (●). (a) $\eta=4.85 \times 10^{-4}$, $k=0.14$. The fitted potential is $\psi_0=155$ mV and the geometrical scaling corresponds to $\sigma=440$ Å. (b) $\eta=1.66 \times 10^{-4}$; all other parameters as for (a).

et al. [1] on suspensions of polystyrene spheres, where the best experimental estimate gives $150 < \psi_0 < 270$ mV. For this data the particle diameter is polydisperse ($\Delta\sigma/\sigma \sim 0.2$), but this should not affect the structure factor, since the effective diameter σ' is generally much larger than the physical diameter σ . We analysed the experimental data as follows. Comparing experimental and theoretical structure factors for one particular sample, we set η and k to the known values and adjusted γ to achieve agreement. We also found that a 4 per cent reduction in σ from the quoted value of 460 ± 23 Å gave better scaling; this change, which may reflect the polydispersity, lies well within the experimental error. The surface potential obtained ($\psi_0 = 155$ mV) lies within the experimental limits, and also agrees with values obtained by computer simulation of the same system [15]. We then kept this value fixed and compared theory with experiment for all other measured states, without any adjustable parameter. A typical example of the good agreement which is achieved is shown in figure 2; transformation of the theoretical curves to $g(r)$ also gives values in agreement with computer simulation [15].

Before concluding, we wish to make two further remarks. (a) In the weak screening limit, the MSA solution of HP [3] requires increasing computational precision because of the wide dynamic range of the coefficients. Under these conditions it is more efficient to use the simpler MSA solution for unscreened charged hard spheres [10] as a reference system, and to treat the weak screening as a perturbation [16]. The presence of the effective hard core allows an application of the 'optimized random phase approximation' (ORPA) [17] to express the structure factor $S(K)$ of the screened system in terms of the structure factor $S_0(K)$ of the unscreened reference system. In practice, $S(K)$ and $S_0(K)$ turn out to be almost indistinguishable for $\kappa a \lesssim 0.5$. Our rescaling prescription could, of course, be equally well applied to the unscreened solution of reference [10]. (b) In a certain range of (γ, η) values corresponding to weak, long ranged interaction (typically $\gamma \sim 1$, $\eta \sim 0.1$), we have noted an interesting effect of the competition between hard core and Coulomb interactions. For sufficiently weak screening (typically $k \lesssim 0.5$), the calculated structure factor exhibits a marked 'shoulder' at small wavenumbers ($K = Q\sigma \simeq 1.0$) resulting from opposing tendencies: a purely hard sphere structure factor would approach its limit

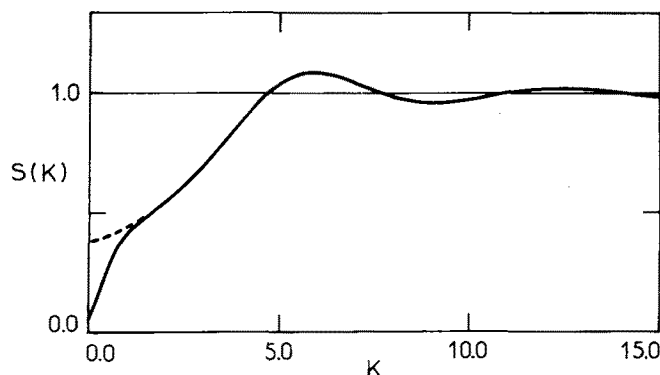


Figure 3. RMSA structure factors for $\gamma=0.2$, $\eta=0.1$ demonstrating the competition between coulombic and excluded volume effects. —, $k=0.2$; ---, $k=1.0$.

$S(Q=0)$ given by the isothermal compressibility (typically $S(0) \simeq 0.4$ for $\eta \simeq 0.1$), while the Coulomb interactions, which become predominant for large spacings (small Q), lead to a vanishing structure factor ($S(Q) \sim Q^2$) in the weak screening limit. A typical case is shown in figure 3.

In summary, we have extended the range of validity of the MSA for screened Coulomb interactions by rescaling to an effective hard core which plays no physical rôle within the Coulomb potential. The resulting theoretical structure factors are sufficiently accurate to provide an unambiguous test of the potential model (1) and a precise estimate of the surface potential ψ_0 of colloidal particles by a confrontation with experimental data.

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